

# CSA5112 - Cryptography and Network Security

## 1. Caesar Cipher

**Aim:** To implement the Caesar Cipher algorithm for encryption and decryption of a given plaintext using a fixed key.

### Algorithm:

1. Start.
2. Read the plaintext and key.
3. For each character in the plaintext:
  - If it is an alphabet, shift it by the key value.
  - Wrap around after 'Z' or 'z'.
4. Display the encrypted text.
5. Repeat the process in reverse for decryption.
6. Stop.

### Code:

```
#include <stdio.h>

#include <string.h>

int main() {

    char text[100];

    int key, i;

    printf("Enter text: ");

    scanf("%s", text);

    printf("Enter key: ");

    scanf("%d", &key);

    printf("Encrypted Text: ");
```

```
for(i = 0; text[i] != '\0'; i++) {  
    if(text[i] >= 'A' && text[i] <= 'Z')  
        printf("%c", ((text[i]-'A'+key)%26)+'A');  
    else if(text[i] >= 'a' && text[i] <= 'z')  
        printf("%c", ((text[i]-'a'+key)%26)+'a');  
}  
printf("\n");  
return 0;  
}
```

### Sample Input:

```
Enter text: HELLO  
Enter key: 3
```

### Sample Output:

```
Encrypted Text: KHOOR
```

### Result:

The Caesar Cipher algorithm was successfully implemented, and the plaintext was encrypted using the given key.